

TransitLens — Abstract · TD2026

TransitLens — Project Abstract

Competition: Transit Data Conference 2026 Data Challenge

Team: TransitLens

Thematic Areas: Advanced data analytics and AI in public transit · Strategic/service planning, performance measurement, and equity analysis · Operations control and incident management · Visualization tools for transit data communication · Practical use of GTFS/GTFS-RT specifications

Abstract (300 words)

Transit agencies operate at the intersection of complex real-time operations and long-term service equity — yet most decision-support tools address one or the other, not both simultaneously. TransitLens is an open-data, real-time intelligence platform for the Toronto Transit Commission (TTC) that unifies operational monitoring, equity analysis, and demand forecasting into a single interactive dashboard.

The platform addresses three core problems facing transit agencies today. First, **operational blind spots**: dispatchers and planners lack real-time visibility into emerging service degradation events such as bus bunching — a phenomenon where vehicles on the same route cluster together, effectively doubling headway for passengers between them.

TransitLens detects bunching live using the Haversine formula applied to TTC GTFS-RT vehicle positions, flagging affected route segments on the map within 30-second polling intervals.

Second, **equity measurement gaps**: neighbourhoods differ dramatically in transit accessibility, yet this disparity is rarely quantified and surfaced in a decision-ready format.

TransitLens computes neighbourhood mobility scores from stop density, average wait time, and income proxies, and generates automated actionable recommendations (e.g., route frequency changes, demand-responsive transit pilots) for the five most underserved areas.

Third, **incident-to-action latency**: safety events from the TTC delay datasets and Toronto Police KSI collision records are currently siloed across separate open data portals. TransitLens fuses these sources into a unified Vision Zero dashboard with automated hotspot detection and route-level incident ranking.

All data is sourced exclusively from open government datasets and public GTFS/GTFS-RT feeds. No personally identifiable information is used or inferred. The prototype is deployed as a web application accessible at <https://transit-lens.vercel.app> and built with React, TanStack Router, Supabase, and React Leaflet.

TransitLens demonstrates that publicly available automated transit data, when properly integrated and visualised, can meaningfully advance service planning, equity analysis, and real-time operations — without requiring proprietary data access.